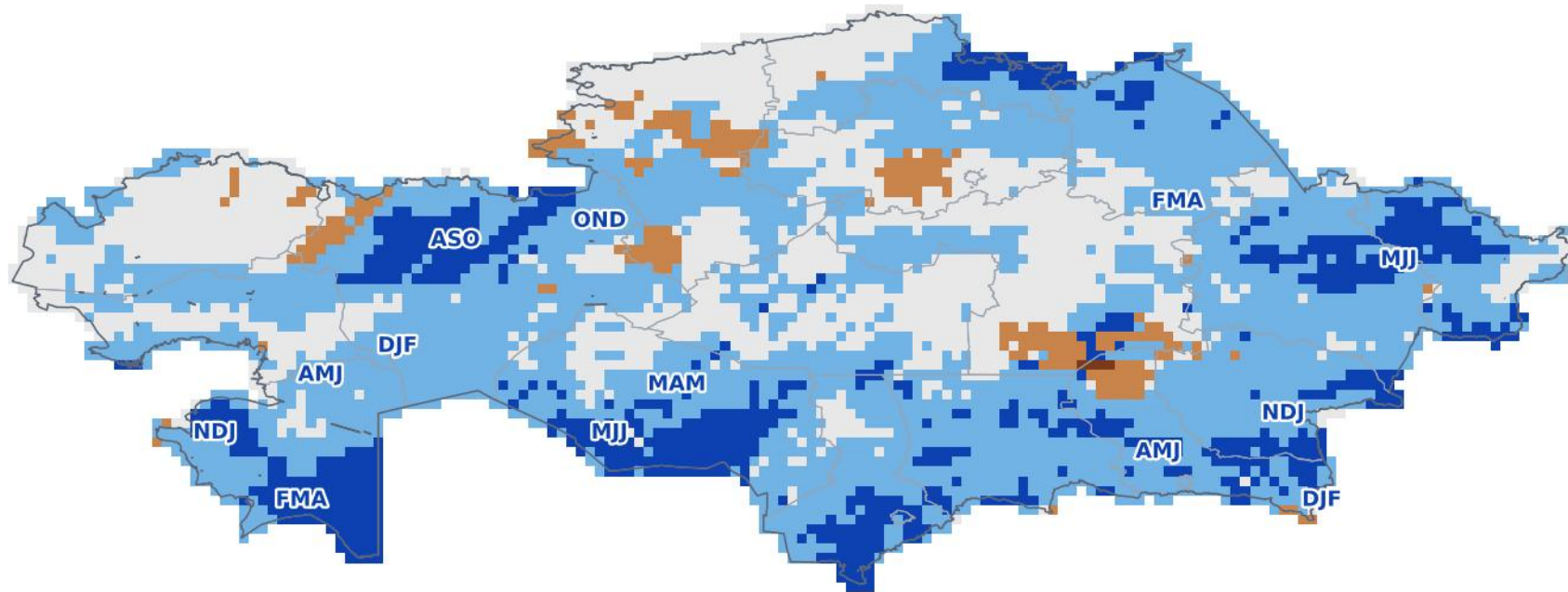


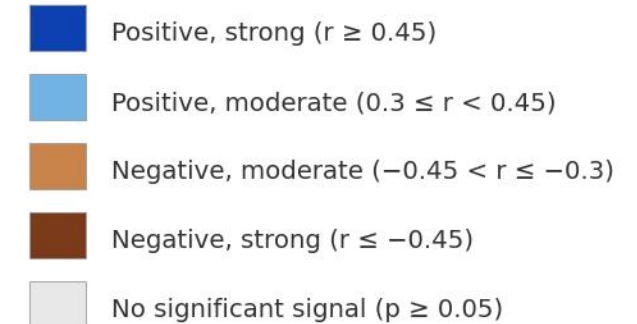
Kazakhstan: how does El Niño generally affect seasonal rainfall?

Pixelwise partial correlation of rainfall with Niño3.4 (ENSO), other climate modes held constant — ERA5 0.25° grid, 1981–2025

Unique ENSO signal — strongest significant trimester per pixel



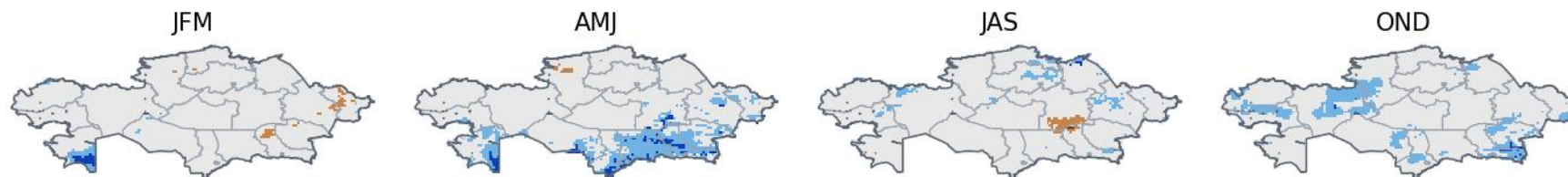
Partial correlation with Niño3.4



Brown = El Niño years tend to be drier than normal in that season; blue = wetter.

Method (the teleconnections page's partial pass):

- Trimester rainfall means per pixel, all 12 overlapping 3-month windows, 1981–2025
- Partial r of rainfall vs Niño3.4, holding DMI (IOD), TNA, TSA and AMM constant
- Each mode at its per-pixel best lag (0–3 mo)
- Two-tailed $p < 0.05$, df adjusted for controls
- All seasons kept (no rainy-season filter)



Data: ERA5 monthly precip (0.25°), NOAA PSL Niño3.4. Bottom row: the four quarters of the year.

Kazakhstan: what seasonal rainfall is predicted this year (August-issued forecast)?

SEAS5 (issued August 2026) accounts for El Niño and every other driver · return-period categories vs the 1981–2025 hindcast · skill-shaded · ADM1

